

PhD Exposé: Multimodal Detection and Analysis of AI-Generated Content

Mohamed Mady

Supervised by: Prof. Dr. Johannes Reschke (OTH Regensburg)

Co-Supervised by: Prof. Dr. Björn Schuller (TUM)

April 25, 2025

Abstract

Rapid advances in artificial intelligence (AI) have revolutionized content creation, enabling the generation of highly realistic synthetic text, images, audio, and videos. These advancements bring significant benefits across various fields, including improved diagnostics in healthcare, personalized learning in education, and automated customer support in business. However, in addition to these benefits, the increasing realism of AI-generated content introduces substantial risks, including misinformation, fraud, and threats to authenticity and integrity.

To effectively counteract these risks, this research proposes a robust multimodal detection framework that integrates computer vision, natural language processing (NLP) and explainable AI (XAI). Current detection methods often function as opaque models, limiting their interpretability and applicability in the real world. By incorporating XAI techniques, such as SHAP, Grad-CAM, and RISE, this approach provides transparency, improves trust, and allows users to understand the rationale behind detection results, essential for practical deployment and ethical compliance.

This multimodal approach combines linguistic, stylometric, statistical, and forensic analyses. Linguistically, I examine features like sentence structure, lexical diversity (e.g., Type-Token Ratio), part-of-speech patterns, passive voice usage, sentiment analysis scores, readability indices, punctuation consistency, and error analysis. Statistically, I use predictive modeling with perplexity, entropy, and burstiness metrics, exploiting the predictability of

AI-generated text compared to human-written content. For images, the approach employs deep learning classifiers alongside forensic methods, analyzing frequency-domain characteristics (spectral artifacts), photo-response non-uniformity (PRNU), error level analysis (ELA), color and texture statistics to detect anomalies typically absent in real photographs.

The study unfolds in three main phases: comprehensive data collection and preprocessing (leveraging datasets such as Hugging Face’s human-vs.-machine text corpus, ArtiFact, and GenImage datasets), multimodal model development, and rigorous evaluation and optimization. By advancing detection methodologies, establishing practical guidelines for identifying AI-generated content, and contributing to ethical and security discussions surrounding AI, this research aims to significantly enhance the verification of content authenticity across critical application domains.

Contents

1	Introduction	4
2	Literature Review	5
2.1	Supervised Deep Learning Approaches	5
2.2	Adversarial Training and Robust Detection	6
2.3	Linguistic and Stylometric Analysis	6
2.4	Statistical Signature Methods	6
2.5	Multimodal Detection Strategies	7
2.6	Statistical and Forensic Image Analysis	7
2.7	Challenges and Limitations	7
3	Methodology	8
3.1	Dataset Construction	8
3.1.1	Textual Data	8
3.1.2	Image Data	9
3.2	Proposed Multimodal Detection Framework	9
3.2.1	Textual Analysis	9
3.2.2	Visual Analysis	10
3.3	Evaluation Metrics	10
3.4	Practical Guidelines and User Awareness	11
4	Challenges and Limitations	11
5	Conclusion	12

1 Introduction

Rapid advances in artificial intelligence (AI) have revolutionized content creation, leading to an exponential increase in AI-generated text, images, and multimedia. Technologies such as Generative Adversarial Networks (GANs) and Transformer-based architectures have enabled the production of highly sophisticated content, often indistinguishable from human-created content. This rapid development presents numerous opportunities across various sectors.

AI-generated content supports innovation and productivity in various fields. Creative industries employ AI for generating compelling text, images, music, and videos, significantly enhancing efficiency and enabling automation of complex tasks. Businesses leverage AI-generated content to improve marketing strategies, automate customer service interactions, and deliver highly personalized recommendations. In academic and professional settings, AI tools assist researchers and professionals by drafting reports, summarizing large volumes of research, and streamlining content creation processes, thereby improving overall productivity and accuracy.

However, alongside these benefits, the increasing sophistication and widespread adoption of AI-generated content raise significant concerns. AI-generated misinformation, deepfake media, and fraudulent documents pose serious risks to journalism, politics, finance, and security. The quality and realism of AI-generated content amplify the potential for deception, complicating efforts to distinguish genuine information from fabricated or manipulated media. In academia, the reliance on AI tools for assignments, research summaries, and scientific papers presents challenges related to plagiarism and threatens academic integrity.

Recent statistics highlight the rapid growth and scale of AI-generated content. Since 2022, more than 15 billion images have been AI-generated, averaging approximately 34 million new images each day, already surpassing the combined total of major stock photography platforms and accelerating further. Similarly, AI-generated text has seen substantial adoption, now accounting for approximately 19.1% of the top 20 search results on Google, a dramatic increase from 2.3% in 2019. Projections indicate that by 2027, over 90 million US adults will depend on AI-generated content as their primary source of online information.

AI-generated content originates from diverse tools and platforms. In text generation, ChatGPT leads with over 50% of user share as of 2023 [8]. Other notable

platforms include Jasper AI (34%), Copy.ai (42%), and Writesonic, each contributing significantly to the growing volume of AI-generated text [17]. For image generation, tools such as Midjourney, DALL·E, and NightCafe each hold approximately 25% of the user base, collectively dominating the AI image generation market [22]. These widely accessible tools have significantly accelerated the proliferation of AI-generated content across digital platforms.

In contrast, traditional indicators of human authorship in academic and professional contexts, such as typographical and grammatical errors, are becoming less reliable due to widespread AI usage. AI-powered tools refine language, correct mistakes, and enhance overall text quality, further complicating efforts to distinguish AI-generated content from human-written material.

Given these challenges, the capability to reliably detect AI-generated content has become essential. Current detection methods, however, typically adopt unimodal approaches, focusing separately on text, images, or audio. To address this limitation, our research proposes a multimodal AI-generated content detection system integrating computer vision, natural language processing (NLP), and explainable AI (XAI) techniques. This multimodal framework aims not only to enhance detection accuracy but also to provide interpretability and transparency, addressing critical ethical and practical considerations in AI content analysis.

2 Literature Review

AI-generated content detection has become increasingly critical due to rapid advancements in generative AI models. Researchers have extensively explored various methodologies, including supervised deep learning, adversarial robustness techniques, zero-shot methods, linguistic and stylometric analyses, statistical signatures, and multimodal detection frameworks. This section summarizes key methodologies, evaluates their effectiveness, and addresses ongoing challenges.

2.1 Supervised Deep Learning Approaches

Early detection methods employed Convolutional Neural Networks (CNNs) to identify AI-generated images and deepfake artifacts. Rössler *et al.* [20] introduced FaceForensics++, a dataset and benchmark leveraging a fine-tuned XceptionNet, which effectively detected facial manipulations but struggled to generalize against

newer generative models introduced since 2019.

For AI-generated text detection, OpenAI developed a RoBERTa-based detector fine-tuned on GPT-2 outputs [26]. Similarly, the Grover model [26], trained on news articles, demonstrated over 92% accuracy in distinguishing real from synthetic texts. However, both models suffered significant accuracy drops when encountering text generated by more advanced, unseen generative techniques developed post-2019.

2.2 Adversarial Training and Robust Detection

To address robustness against adversarial modifications, Durall *et al.* [5] exploited frequency-domain anomalies typically present in GAN-generated images, outperforming traditional pixel-based CNNs. Nonetheless, adversarial methods capable of slight spectral alterations (e.g., smoothing or adding subtle noise) significantly reduced detection accuracy.

In the text domain, DetectGPT [16] offered a zero-shot detection method relying on statistical anomalies derived from probability curvature without the need for labeled training data. Although effective and generalizable, DetectGPT requires direct access to a language model’s probability distributions, limiting practical deployment.

2.3 Linguistic and Stylometric Analysis

Linguistic and stylometric methods provide additional cues for detecting AI-generated text by evaluating metrics such as vocabulary richness, sentence structure consistency, part-of-speech distributions, average sentence length, readability indices (e.g., Flesch-Kincaid), lexical diversity metrics (Type-Token Ratio, Mean Segmental TTR, MTLD), passive voice usage, sentiment analysis scores, and error analysis (spelling or grammatical inconsistencies). Purely linguistic methods, however, often fail if AI-generated texts undergo significant human editing.

2.4 Statistical Signature Methods

Perplexity, entropy, and burstiness metrics offer statistical approaches for AI-generated text detection. Perplexity measures the predictability of textual sequences, with AI-generated texts generally exhibiting lower average perplexity due to predictable linguistic patterns. Burstiness, measuring variance in sentence-level perplexity, and predictive entropy (uncertainty of next-token predictions) further differentiate AI

from human-generated texts. However, these methods require suitable reference models aligned with generative model sophistication, making them sensitive to newer, advanced AI models.

2.5 Multimodal Detection Strategies

Multimodal approaches combining textual and visual analyses have shown increased detection robustness. Shao *et al.* [21] developed HAMMER, a hierarchical transformer that identifies inconsistencies between text descriptions and corresponding images, significantly enhancing detection accuracy over unimodal models, particularly in fake news scenarios.

Cozzolino *et al.* [4] proposed a multimodal detection framework integrating entropy-based image analysis and linguistic features, achieving state-of-the-art zero-shot detection performance. Such approaches underscore multimodal integration’s potential in tackling sophisticated generative models.

2.6 Statistical and Forensic Image Analysis

Beyond deep learning, researchers explored forensic techniques such as Photo Response Non-Uniformity (PRNU), Error Level Analysis (ELA), and frequency-domain analyses. PRNU leverages camera-specific sensor noise absent in AI-generated images, while ELA detects inconsistent compression artifacts. Frequency-domain analyses identify GAN-specific spectral anomalies like checkerboard patterns, though these methods remain vulnerable to post-processing attacks.

Color and texture statistics, including pixel intensity co-occurrence matrices and abnormal color correlations, have also proven effective for earlier GAN models but are less reliable against advanced generative models and refined post-processing.

2.7 Challenges and Limitations

Despite notable advancements, AI-generated content detection encounters several critical challenges:

- **Adversarial Robustness:** Generative models rapidly evolve, introducing sophisticated adversarial attacks capable of evading established detection mechanisms [5, 23].

- **Generalization Issues:** Detection models frequently exhibit poor generalization to newer generative techniques and models introduced post-2019, highlighting the need for adaptive detection methods [11, 7].
- **Computational Scalability:** Many high-performing models require extensive computational resources, limiting real-time applicability and large-scale deployment [2].
- **Ethical Considerations:** Excessive dependence on AI-detection technologies poses risks of censorship and misuse, potentially suppressing legitimate content [12].

Advancing AI-generated content detection requires ongoing development of multimodal and hybrid methods integrating multiple signals (textual, visual, statistical, forensic) and addressing robustness, generalization, scalability, and ethical concerns.

3 Methodology

The primary challenge identified in existing research is the generalization and robustness of AI-generated content detection models. Current methodologies often demonstrate limited adaptability to novel generative techniques due to overfitting specific datasets. Addressing this, the proposed approach emphasizes dataset diversity, multimodal analysis, interpretability, and comprehensive evaluation.

3.1 Dataset Construction

A specialized dataset focused on academic, and professional documents and general diverse images will be utilized to ensure diversity and broad applicability across multiple content domains.

3.1.1 Textual Data

Authentic human-written documents will be collected from:

- **OPUS Repository (OTH Regensburg):** Theses and research papers submitted in or before 2022.
- **Technical University of Munich (TUM):** Archived academic reports and theses from the same period.

For each authentic document, a corresponding AI-generated version will be created using widely used generative language models such as GPT (OpenAI), Gemini (Google), DeepSeek, and others. This will result in a balanced dataset including:

- Fully AI-generated reports.
- Original human-authored reports.
- Hybrid versions (human-written content with AI-assisted sections).

3.1.2 Image Data

The visual dataset will focus on detecting synthetic images across diverse domains, leveraging the following large-scale benchmark datasets:

- **ArtiFact** (ICIP 2023) [18]: A large-scale dataset for generalizable fake detection, containing approximately 1.53 million AI-generated images from various generative models and 0.96 million real images sourced from COCO and FFHQ.
- **GenImage** [10]: A comprehensive dataset with over 1.35 million AI-generated images (produced by models such as Stable Diffusion, MidJourney, and BigGAN) paired with an equal number of real images drawn from ImageNet.

Visual comparisons between real and synthetic examples will be included to highlight detection difficulty and justify the necessity of this research.

3.2 Proposed Multimodal Detection Framework

The proposed detection framework combines textual and visual analyses, integrating explainability methods to ensure interpretability.

3.2.1 Textual Analysis

Textual analysis will incorporate:

- **Stylometric and Linguistic Features:** Including sentence length variability, passive voice usage, lexical diversity metrics (Type-Token Ratio, MTLD, MSTTR, Herdan’s C, Vocd), sentiment scores, readability indices (e.g., Flesch-Kincaid), punctuation usage, and grammatical consistency [16].
- **Transformer-Based Models:** RoBERTa, DeBERTa, and other pre-trained transformer models fine-tuned on the constructed dataset.

- **Statistical Signature Analysis:** Perplexity, burstiness, and entropy analysis using reference language models (e.g., GPT-2 and GPT-3).
- **AI Probability Scores:** Providing a graded scale indicating the likelihood of AI-generated text.

3.2.2 Visual Analysis

The visual analysis approach integrates:

- **Frequency Domain Analysis:** Detection of spectral anomalies using Fourier transforms and spectral CNN methods [5].
- **CNN-Based Detection:** Utilizing architectures like EfficientNet and ResNet adapted specifically for scientific and medical imagery.
- **Photo Response Non-Uniformity (PRNU) Analysis:** Detecting the absence of sensor noise characteristic of authentic camera images [4].
- **Error Level Analysis (ELA):** Highlighting recompression artifacts indicative of AI-generated images.
- **Texture and Color Analysis:** Co-occurrence matrices and color channel anomalies detection.
- **Multimodal Fusion Techniques:** Transformer-based fusion architectures like ViLBERT and other cross-modal integration methods [14, 21].

3.3 Evaluation Metrics

The proposed model’s performance will be rigorously evaluated using:

- **Accuracy, Precision, Recall, F1-Score:** Standard classification metrics.
- **Area Under Receiver Operating Characteristic Curve (AUROC):** Robustness across thresholds.
- **Equal Error Rate (EER):** Evaluating false positives and negatives.
- **Explainability Metrics:** Ensuring the interpretability and reliability of explanations provided by SHAP, LIME, Grad-CAM, and RISE.
- **Cross-Dataset Generalization:** Testing generalizability with unseen datasets.
- **Probability Score Calibration:** Ensuring accurate probabilistic predictions for AI-generated content.

3.4 Practical Guidelines and User Awareness

Beyond automated detection, guidelines for manual identification of AI-generated content will be developed to support human evaluators in recognizing potential synthetic content. This component addresses ethical considerations and practical deployment scenarios where automated detection might not suffice.

The outlined methodology emphasizes robust multimodal detection, comprehensive interpretability, and extensive evaluation, addressing both technical and ethical challenges in distinguishing authentic from AI-generated content.

4 Challenges and Limitations

Although the proposed multimodal detection framework addresses several shortcomings in existing methods, significant challenges and limitations remain:

- **Data Availability and Bias:** Obtaining a diverse, high-quality corpus of authentic, human-authored documents (especially pre-2022) can be challenging. Limited diversity or biases in collected documents could negatively affect the generalizability of the detection model.
- **Evolution of Generative Models:** AI-generated content technologies evolve rapidly, continuously producing more sophisticated outputs. Keeping detection methodologies up-to-date requires constant adaptation, extensive re-training, and rigorous cross-dataset validation to ensure continued robustness against newly emerging generative methods [5, 11].
- **Changes in Human Writing Style:** The widespread adoption of AI tools like ChatGPT influences human writing, potentially blurring traditional linguistic and stylistic differences between AI-generated and human-written texts. Over time, this convergence could significantly complicate detection, necessitating continuous model updates and new feature development to distinguish increasingly subtle patterns.
- **Computational Resource Constraints:** Multimodal and ensemble methods, especially those involving transformer-based architectures and frequency-domain image analyses, require substantial computational resources. Real-time or large-scale applications may become impractical without optimization strategies, such as model distillation or efficient hardware deployments [2].

- **Explainability and Ethical Implications:** Balancing accuracy and interpretability remains critical. Black-box approaches undermine user trust, while highly interpretable methods (XAI) risk increased computational complexity. Additionally, false positives could lead to misclassifying genuine content, raising ethical concerns about censorship, privacy, and trustworthiness [12].
- **Adversarial Robustness:** Generative models can intentionally incorporate counter-forensic techniques, like frequency smoothing or adversarial perturbations, to evade detection. Developing robust methods resilient to adversarial manipulation is crucial to maintaining effective long-term detection [23].
- **Real-World Deployment and Practicality:** Ensuring that detection models perform reliably under realistic conditions, such as compression artifacts, resizing, and varied quality, poses additional challenges. Benchmarks indicate reduced effectiveness under such practical scenarios, necessitating comprehensive evaluation protocols to mimic real-world usage closely [21, 7].

5 Conclusion

The rapid advancements in AI-generated content technologies present significant benefits across various sectors, including healthcare, education, business, and academia. However, these developments also introduce considerable challenges, such as misinformation, academic dishonesty, and the potential misuse of synthetic medical documentation. To address these concerns, this research proposes a robust, multimodal AI-generated content detection framework that integrates both textual and visual analysis while ensuring interpretability through Explainable AI (XAI) methods.

The study emphasizes constructing a diverse, high-quality dataset comprising authentic human-generated and AI-synthesized academic texts, medical reports, and associated visual content, particularly targeting reports and documents from before widespread AI adoption (pre-2022). This targeted data collection supports generalizability, robustness, and mitigates biases in detection performance.

Our multimodal detection approach combines linguistic and stylometric text features (including lexical diversity, perplexity, and readability indices), frequency-domain image analysis, and CNN-based visual detection. We employ transformer-based models like RoBERTa and EfficientNet, enhanced by XAI techniques (SHAP, Grad-CAM, LIME) to provide transparency in predictions. Additionally, intro-

ducing an AI Probability Score allows for nuanced assessments rather than binary classifications, aiding end-users in interpreting results with greater confidence.

Several challenges remain critical, notably data availability constraints, rapid evolution in generative technologies, changes in human writing styles influenced by AI tools, computational limitations, ethical implications, and adversarial vulnerabilities. Addressing these will involve continuous model updates, adversarial training, careful ethical considerations, and extensive real-world validation.

Ultimately, this research aims to produce practical and reliable tools for authenticating academic submissions, such as student-generated theses and research papers, as well as medical documentation, safeguarding integrity in educational institutions, healthcare systems, and professional environments. Future work will focus on enhancing adaptability to emerging generative techniques, improving computational efficiency for large-scale deployment, and providing clear guidelines for users to identify AI-generated content reliably.

References

- [1] B. Bayar and M. C. Stamm, “A Deep Learning Approach to Universal Image Manipulation Detection Using a New Convolutional Layer,” in *Proc. ACM IH&MMSec*, 2016.
- [2] R. Chesney and D. Citron, “Deepfakes and the New Disinformation War,” in *Foreign Affairs*, 2019.
- [3] C2PA, “Content Provenance and Authenticity Technical Specifications,” Coalition for Content Provenance and Authenticity, 2023. [Online]. Available: <https://c2pa.org/specifications/specifications/1.3/>
- [4] D. Cozzolino, J. Thies, and M. Nießner, “Zero-Shot Detection of AI-Generated Images,” in *ECCV*, 2024.
- [5] R. Durall, M. Keuper, and J. Keuper, “Watch Your Up-Convolution: CNN Based Generative Deep Fake Detection via Frequency Analysis,” in *Proc. CVPR Workshops*, 2020.
- [6] D. Epstein, D. Cozzolino, and L. Verdoliva, “Learning to Detect AI-Synthesized Images from a Timeline of Generative Models,” in *Proc. CVPR*, 2023.

- [7] T. Fagni, F. Falchi, D. Gambini, and M. Tesconi, “TweepFake: About detecting deepfake tweets,” in *PLOS ONE*, vol. 16, no. 5, e0251415, 2021.
- [8] Generative AI Top 150: The World’s Most Used AI Tools. Retrieved from <https://www.flexos.work/learn/generative-ai-top-150>
- [9] J. Frank, T. Eisenhofer, L. Schönherr, A. Fischer, J. Kolossa, and T. Holz, “Leveraging Frequency Analysis for Deep Fake Image Recognition,” *arXiv preprint arXiv:2003.08685*, 2020.
- [10] W. Zhang, Y. Jiang, L. Xu, and S. Wang, “GenImage: A Large-Scale Dataset of Generative Images with Real Counterparts,” *arXiv preprint arXiv:2305.18288*, 2023.
- [11] H. Khalid, N. Rahman, and M. Ahmed, “FakeAVCeleb: A Novel Audio-Video Multimodal Deepfake Dataset,” in *ICASSP*, 2021.
- [12] J. Kirchenbauer, F. Albert, J. Geiping, M. Moeller, and A. Tomkins, “A Watermark for Large Language Models,” *arXiv preprint arXiv:2301.10226*, 2023.
- [13] X. Li, Z. Sun, X. Shen, A. B. Yao, and W. Lu, “DocBank: A Benchmark Dataset for Document Layout Analysis,” in *Proc. COLING*, 2020.
- [14] J. Lu, D. Batra, D. Parikh, and S. Lee, “ViLBERT: Pretraining Task-Agnostic Visiolinguistic Representations for Vision-and-Language Tasks,” in *NeurIPS*, 2019.
- [15] D. S. W. Paul et al., “MedNIST Dataset: Medical Imaging for Deep Learning Applications,” 2019. Available: <https://github.com/Project-MONAI/MONAI/tree/main/examples>
- [16] E. Mitchell, J. Lee, A. Khazatsky, C. Liang, and S. Manning, “DetectGPT: Zero-Shot Machine-Generated Text Detection using Probability Curvature,” in *ICML*, 2023.
- [17] Important Generative AI Statistics (Sept 2024). Retrieved from <https://nerdynav.com/generative-ai-statistics/>
- [18] T. H. Rahman, M. S. Jain, H. Kwon, S. Dutta, and A. K. Roy-Chowdhury, “ArtiFact: A Large-Scale Benchmark Dataset for Generalizable Deepfake Detection,” in *Proc. IEEE Int. Conf. Image Process. (ICIP)*, 2023.

- [19] M. Ribeiro, S. Singh, and C. Guestrin, “Why Should I Trust You? Explaining the Predictions of Any Classifier,” in *Proc. KDD*, 2016.
- [20] A. Rössler, D. Cozzolino, L. Verdoliva, C. Riess, J. Thies, and M. Nießner, “FaceForensics++: Learning to detect manipulated facial images,” in *Proc. ICCV*, 2019.
- [21] R. Shao, L. Cui, and W. Huang, “DGM4: Detecting and Grounding Multi-Modal Media Manipulation,” in *CVPR*, 2023.
- [22] Global generative AI image tool market share 2023. Retrieved from <https://www.statista.com/forecasts/1423989/world-generative-ai-image-tool-market-share>
- [23] X. Yang, H. Li, and S. Lyu, “On the Continuous Evolution of Deepfake Generators: Analysis and Countermeasures,” in *CVPR*, 2021.
- [24] N. Yu, L. Davis, and M. Fritz, “Attributing Fake Images to GANs: Learning and Analyzing GAN Fingerprints,” in *NeurIPS*, 2019.
- [25] X. Zhong, J. Tang, and A. H. M. Peterson, “PubLayNet: Largest Dataset Ever for Document Layout Analysis,” *arXiv preprint arXiv:1908.07836*, 2019.
- [26] R. Zellers, A. Holtzman, H. Rashkin, Y. Bisk, and Y. Choi, “Defending Against Neural Fake News,” in *NeurIPS*, 2019.